

Waleed Ahmed

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RESEARCH PROJECTS

02/09/2024 – 01/01/2025
Structural Composite Supercapacitors for Multifunctional Engineering Applications

- Studied the development of structural supercapacitors that combine **mechanical load-bearing capability with energy storage functionality.**
- Examined the role of **carbon nanotube (CNT) reinforced composites** in improving electrical conductivity and structural performance.
- Evaluated the importance of lightweight multifunctional materials for **electric vehicles and aerospace structures**.
- Reviewed how integrated energy-storing structures can contribute to future **efficient and compact power solutions**.

01/10/2024 – 01/11/2024
Recent Developments in Lithium-Ion Battery Materials

- Reviewed emerging technologies in lithium batteries including **solid-state electrolytes and silicon-based anodes**.
- Analyzed improvements in **battery safety, cycle life, and energy storage capacity**.
- Investigated the contribution of nanomaterials toward improving battery efficiency.
- Studied material challenges related to performance degradation and thermal stability.

02/09/2024 – 03/01/2025
ETFE-Based Protective Layer for Solar Panel Applications

- Investigated the use of Ethylene Tetrafluoroethylene (ETFE) as an alternative protective layer for photovoltaic solar panels.
- Evaluated ETFE for its high crack resistance, mechanical strength, and environmental durability compared to conventional solar panel encapsulation materials.
- Studied the potential of ETFE to enhance panel lifespan by improving resistance to mechanical damage and environmental degradation.
- Analyzed the feasibility of replacing traditional top protective layers with ETFE to improve performance and reliability of solar modules.

04/09/2023 – 01/01/2024
Development of a Micro Universal Testing Machine for Thin Material Characterization

- Contributed to the design of a compact **Universal Testing Machine (UTM)** for testing thin films and fiber materials.
- Utilized a **10 N load cell** for precise measurement of small mechanical loads.
- Integrated precision gripping mechanisms to prevent specimen damage.
- Assisted in implementing a **data acquisition system** for accurate stress-strain analysis.

01/01/2024 – 01/05/2024
Sustainable Biomass-Derived Materials for Advanced Engineering Uses

- Investigated renewable biomass as a source for developing **biodegradable engineering materials**.
- Explored applications in **sustainable packaging, green construction materials, and energy devices**.
- Studied the environmental advantages of bio-based materials compared to conventional polymers.

EDUCATION AND TRAINING

12/10/2020 – 01/06/2025 Islamabad, Pakistan
BACHELOR OF SCIENCE IN METALLURGY AND MATERIALS ENGINEERING National University of Sciences and Technology (NUST)

- **CGPA:** 3.15 / 4.00
- Ranked **353** in QS World Ranking and **144th** in QS Subject Ranking.
- **Relevant Coursework:** Nanomaterials, Advanced Materials, Composite Materials, Kinetics of Thermodynamics and Phase Transformations.

Website https://nust.edu.pk/ | **Level in EQF** EQF level 6

WORK EXPERIENCE

MATERIALS ENGINEERING INTERN – SCIENTIFIC & ENGINEERING SERVICES DIRECTORATE (SES) – 01/07/2024 – 09/08/2024 – ISLAMABAD, PAKISTAN

- **Supervisors:** Mr. Abdul Jawad (Manager Training) & Mr. Salik Qadeer Butt (Senior Director).
- **Key Responsibilities:**
- Executed advanced welding techniques including TIG and Gas Welding.
- Conducted Non-Destructive Testing (NDT) inspections to detect defects in industrial welds, directly applicable to particle detector maintenance.
- Preparing and submitting reports after every specimen tested.

UNDERGRADUATE THESIS – SCHOOL OF CHEMICAL AND MATERIALS ENGINEERING, NUST – 02/09/2024 – 06/06/2025 – ISLAMABAD, PAKISTAN

- Developed a QSAR-based machine learning model to identify potential drug candidates targeting glucocorticoid receptors for luminal breast cancer treatment.
- Performed molecular descriptor calculation, feature selection, and PCA using MOE, Python, and Google Colab for predictive modeling.
- Applied molecular docking techniques to evaluate protein–drug interactions and validate predicted compounds.
- Contributed to computational drug discovery by integrating materials science, nanotechnology, and AI-based screening approaches.

VOLUNTEERING

Volunteer & Philanthropy Work

- **09/2023 – 09/2024:** Member of **NUST Sustainability Office**; promoted green campus initiatives.
- **2022 – 2025:** Volunteer Educator at local orphanages in Islamabad; taught basic sciences and mathematics.
- **2023:** Organizer, Plantation Drive at Orphanage.
- **2022 – 2025:** Graphic Designer for the School of Chemical and Materials Engineering (SCME).

SKILLS

Lab Equipment

Vickers Hardness Testing | Brinnel Hardness Testing | Rockwell Hardness Testing | Lathe Machine Operation | FTIR | XRD | UTM | Charpy Impact testing | SEM | Optical Microscopy | Spectroscopy | Fatigue and Creep Test | DSC | XPS | TGA | EDX

Simulation & Design

Molecular Operating Environment (MOE) | LaTeX

Development

MERN stack web developement | Git

Data Science

Num Py | Google Colab | QSAR Modeling | Sickit-learn | Python | Pandas

Fabrication

Sol-Gel Synthesis | Heat Treatment | Welding (TIG/Gas)

CONFERENCES AND SEMINARS

05/05/2025 – 08/05/2025 NICE Seminar Hall, NUST, Islamabad, Pakistan
National University of Sciences and Technology (NUST)

- **Conference:** Attended The 4th ASEAN-Pakistan Conference on Materials Science (APCoMS), 2025.

HONOURS AND AWARDS

Awards

- **Grade A** in Final Year Design Project (2025).
- **Winner**, Inter-department Football Tournament (2024).
- **Winner**, Inter-college Football Tournament (2018).
- **Participant**, HRCA English & Science Quiz (2017-18).